

An Archaeological Inference Problem

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Monday, February 24, 2025

About the Paper

"An Archaeological Inference Problem", Published by the *Journal of the American Statistical Association*, 1988

- **Goal:** Estimate the start and end dates of ceramic phases of artifacts from an Iron Age hill fort by creating a statistical model
- **Challenges:** There are difficulties with mapping radiocarbon dates to actual calendar dates and with getting accurate radiocarbon dates
- **Why:** Precise dates are needed when using artifacts to create chronologies and establish timelines of cultural developments

Archaeological Context

- Site: Danebury Iron Age hill fort, England
- Determining Ceramic Phases:
 - Archaeologist collected 51 samples of wood charcoal, 9 samples of grain, and 10 samples of animal bone
 - Each sample is associated with a pottery shard that was excavated from the same location
 - Archaeologists classify the pottery fragments into one of four phases, called Ceramic Phases 1-4



The Data

Table 1. Ceramic Phase Data

Ceramic Phase 1 (11 observations)		
2,530 $\pm$ 110	2,370 $\pm$ 80	2,460 $\pm$ 60
2,420 $\pm$ 80	2,440 $\pm$ 70	2,210 $\pm$ 60
2,160 $\pm$ 80	2,330 $\pm$ 60	2,450 $\pm$ 80
2,770 $\pm$ 70	2,300 $\pm$ 100	
Ceramic Phase 2 (9 observations)		
2,290 $\pm$ 60	2,270 $\pm$ 90	2,120 $\pm$ 70
2,330 $\pm$ 90	2,140 $\pm$ 80	2,580 $\pm$ 80
2,340 $\pm$ 100	2,300 $\pm$ 90	2,180 $\pm$ 90
Ceramic Phase 3 (17 observations)		
2,230 $\pm$ 70	2,210 $\pm$ 70	2,090 $\pm$ 90
2,060 $\pm$ 80	2,090 $\pm$ 70	2,110 $\pm$ 70
2,210 $\pm$ 70	2,210 $\pm$ 70	2,250 $\pm$ 70
2,120 $\pm$ 80	2,470 $\pm$ 90	2,280 $\pm$ 70
2,380 $\pm$ 90	2,520 $\pm$ 80	2,300 $\pm$ 70
2,220 $\pm$ 90	2,330 $\pm$ 70	
Ceramic Phase 4 (23 observations)		
2,140 $\pm$ 80	2,160 $\pm$ 70	1,980 $\pm$ 80
2,030 $\pm$ 70	2,200 $\pm$ 80	2,260 $\pm$ 80
2,100 $\pm$ 90	2,300 $\pm$ 70	2,090 $\pm$ 60
2,110 $\pm$ 80	2,060 $\pm$ 70	2,120 $\pm$ 70
2,060 $\pm$ 60	2,170 $\pm$ 70	2,000 $\pm$ 70
1,990 $\pm$ 70	2,370 $\pm$ 70	2,130 $\pm$ 60
2,170 $\pm$ 70	2,120 $\pm$ 70	1,900 $\pm$ 60
2,040 $\pm$ 70	2,150 $\pm$ 70	

Challenges of Radiocarbon Dating

Radiocarbon Dating

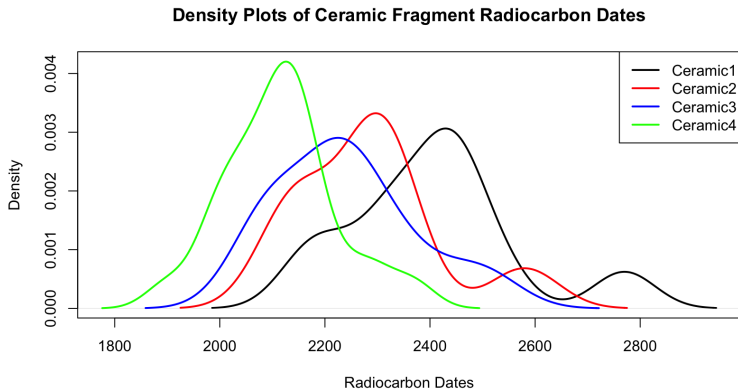
The remaining carbon-14 in an organism is measured since we know its rate of decay (half life of 5,730 years)

- The amount of carbon-14 varies in the atmosphere so the given level of carbon-14 could correspond to an incorrect calendar date
- Uncertainty in radiocarbon dating

Calibration curves are used to account for fluctuations in atmospheric carbon-14 levels and provide more accurate calendar dates.

- To create calibration curves, measure carbon-14 in samples with known ages, such as tree rings

Illustrating the Issue

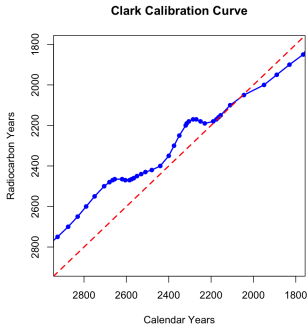


Since the boundaries between the ceramic phases are not clear-cut, we need to use advanced statistical methods to estimate the boundary dates.

Calibration Curve

The curve in this paper is based on dendrochronology:

- Scientists measure the carbon-14 content in tree rings of known age to create the curve
- Used to estimate the calendar dates corresponding to the radiocarbon dates of ceramic artifacts



Without calibration, radiocarbon dating could be off by hundreds of years.

Overview of the Model

Instead of assuming that each fragment definitely belongs to a single phase based on its raw radiocarbon date, the authors use likelihood functions to model the probability that a given date comes from a particular phase.

- Uses statistical distributions to estimate the boundaries of ceramic phases,

$$\alpha^T = (\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5), \quad \alpha_1 > \alpha_2 > \alpha_3 > \alpha_4 > \alpha_5.$$

$$p(\theta|j, \alpha) = p(\theta|\alpha_j, \alpha_{j+1}), \quad \alpha_j > \theta \geq \alpha_{j+1}.$$

- Relates radiocarbon measurements to actual calendar dates via calibration curves.

Variation and Unknowns in the Model

Sources of Variation:

- radiocarbon dating uncertainty
- calibration curve uncertainty
- phase boundaries overlap
- sampling variation

Unknown Variables:

- **Phase Boundaries** (α_j): Start and end calendar dates for each ceramic phase
- **True Calendar Dates** (θ_j): Actual historical date of when the artifact was created → lies within its assigned ceramic phase
- **Radiocarbon Date** μ : Radiocarbon date is related to the production date via the calibration curve $\mu = \mu(\theta)$

Modeling Radiocarbon Dates

Normal Distribution

A probability distribution with a bell-shaped curve symmetric about the mean μ . This means that values near the mean occur more frequently than values further from the mean.

$$p(x|\mu, s) = \frac{1}{\sqrt{2\pi}s} \exp\left(-\frac{(x - \mu)^2}{2s^2}\right)$$

- x = the observed radiocarbon date
- μ = actual radiocarbon date
- s = standard deviation

In the context of radiocarbon dating, we assume that measured radiocarbon date (x) follow a normal distribution centered on the "true" radiocarbon date $\mu(\theta)$.

Modeling Calendar Dates

Uniform Distribution

A probability distribution where all values within a given interval are equally likely. It had a constant probability across its range.

$$p(\theta|\alpha_j, \alpha_{j+1}) = \frac{1}{\alpha_j - \alpha_{j+1}}, \text{ if } \alpha_j \geq \theta > \alpha_{j+1}$$

- j = ceramic phase
- θ = actual date of manufacture of the original pot (calendar years, bp)

In the context of radiocarbon dating, we assume that calendar dates within a given ceramic phase are uniformly distributed.

Implementing the Calibration Curve

A calibration curve $\mu(\theta)$ maps radiocarbon dates to calendar dates and is modeled as a piecewise linear function:

Knots

Breakpoints in a piecewise function, where the slope transitions between different linear segments.

A set of knots is defined at $K + 1$ points $((t_k, m_k), k = 0, 1, \dots, K)$, where t_k are calendar years (θ) and m_k are radiocarbon years (μ). Between two adjacent knots, the relationship is linear:

$$\mu(\theta) = a_k + b_k(\theta), \quad t_{k-1} < \theta \leq t_k,$$

Where a_k is the intercept and b_k is the slope.

Constructing the Likelihood Function

We want to compute the probability of observing radiocarbon dates x given the ceramic phase j .

$$p(x|s, j, \alpha) = \int_{\alpha_{j+1}}^{\alpha_j} p(x|\mu(\theta), s)p(\theta|\alpha_j, \alpha_{j+1})d\theta$$

The likelihood for each observation (x, s, j) integrates over all possible calendar dates θ within the ceramic phase $(\alpha_j, \alpha_{j+1}]$.

Next Steps

- Derive the log-likelihood function
- Replicate their likelihood function calculations
- Examine how Bayesian inference techniques are used to estimate the ceramic phase boundaries (posterior density)



Excavations at Danebury